

Cloud Forensic Investigation: Analysis of AWS Credential Compromise and Data Exfiltration

Objective

This project investigates a simulated AWS cloud compromise using CloudTrail logs. The goal is to identify how the attacker gained access, escalated privileges, exfiltrated sensitive data, and maintained persistence within the environment.

Tools Used

VS Code : log analysis and JSON parsing

Base64 Decoder (CyberChef) : for decoding obfuscated data

Methodology

The investigation was conducted in the following steps:

Log Familiarization

Reviewed CloudTrail logs to understand structure (eventTime, eventName, sourceIPAddress, userIdentity)

Noise vs Signal Filtering

Identified normal AWS internal activity (internal IPs, system agents)

Focused on external IPs and suspicious API calls

Credential Tracking

Traced repeated use of Access Key IDs across events

Timeline Construction

Ordered events chronologically to track attacker progression

Data Decoding

Extracted and decoded Base64-encoded strings found in logs

Behavioral Analysis

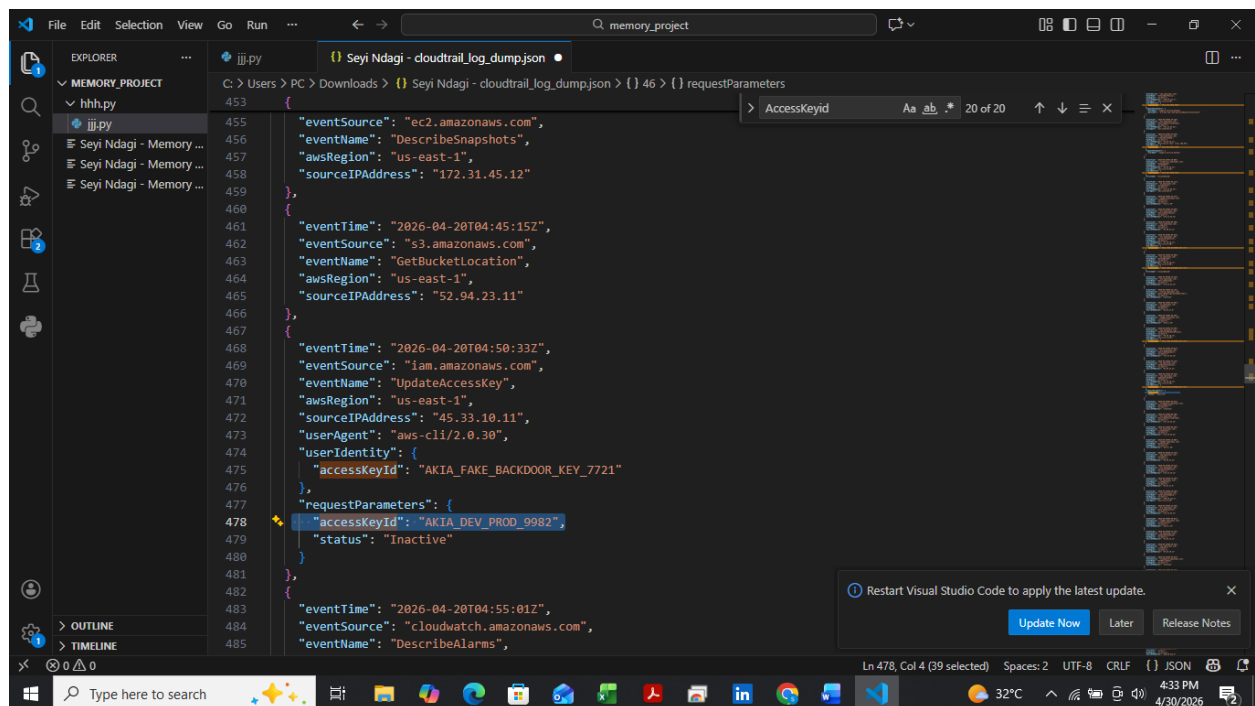
Identified privilege escalation, persistence mechanisms, and data exfiltration

Investigation Findings

1. Credential Compromise

The attacker used the following compromised AWS access key:

AKIA_DEV_PROD_9982



```
455 {
456   "eventSource": "ec2.amazonaws.com",
457   "eventName": "DescribeSnapshots",
458   "awsRegion": "us-east-1",
459   "sourceIpAddress": "172.31.45.12"
460 },
461 {
462   "eventTime": "2026-04-20T04:45:15Z",
463   "eventSource": "s3.amazonaws.com",
464   "eventName": "GetBucketLocation",
465   "awsRegion": "us-east-1",
466   "sourceIpAddress": "52.94.23.11"
467 },
468 {
469   "eventTime": "2026-04-20T04:50:33Z",
470   "eventSource": "iam.amazonaws.com",
471   "eventName": "UpdateAccessKey",
472   "awsRegion": "us-east-1",
473   "sourceIpAddress": "45.33.10.11",
474   "userAgent": "aws-cli/2.0.30",
475   "userIdentity": {
476     "accessKeyId": "AKIA_FAKE_BACKDOOR_KEY_7721"
477   },
478   "requestParameters": {
479     "accessKeyId": "AKIA_DEV_PROD_9982",
480     "status": "Inactive"
481   },
482 },
483 {
484   "eventTime": "2026-04-20T04:55:01Z",
485   "eventSource": "cloudwatch.amazonaws.com",
486   "eventName": "DescribeAlarms",
```

This key was repeatedly observed performing reconnaissance and administrative actions, confirming it as the primary compromised credential.

2. Attacker IP Analysis

Initial Attacker IP: 190.45.112.3

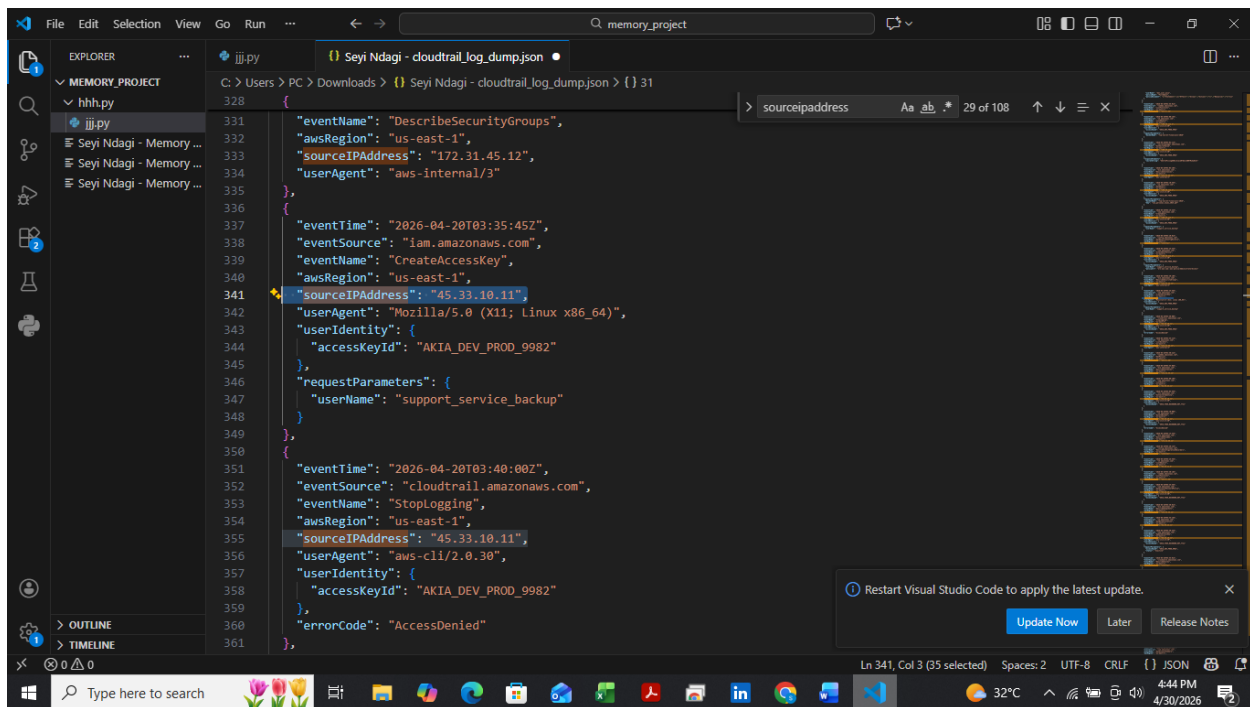
```
26 {
31   "sourceIPAddress": "10.0.1.55",
32   "userAgent": "Boto3/1.26.1",
33   "userIdentity": {
34     "type": "AssumedRole",
35     "principalId": "AROA:LambdaExec"
36   }
37 },
38 {
39   "eventTime": "2026-04-20T01:15:33Z",
40   "eventSource": "sts.amazonaws.com",
41   "eventName": "GetCallerIdentity",
42   "awsRegion": "us-east-1",
43   "sourceIPAddress": "190.45.112.3",
44   "userAgent": "aws-cli/2.0.30",
45   "userIdentity": {
46     "type": "IAMUser",
47     "accessKeyId": "AKIA_DEV_PROD_9982"
48   }
49 },
50 {
51   "eventTime": "2026-04-20T01:20:10Z",
52   "eventSource": "ec2.amazonaws.com",
53   "eventName": "DescribeSecurityGroups",
54   "awsRegion": "us-east-1",
55   "sourceIPAddress": "172.31.45.12",
56   "userAgent": "aws-internal/3"
57 },
58 {
59   "eventTime": "2026-04-20T01:23:45Z",
60   "eventSource": "s3.amazonaws.com",
61   "eventName": "PutObject",
```

First observed during identity verification (GetCallerIdentity)

Used for reconnaissance and privilege escalation

Secondary IP (Persistence Phase):

45.33.10.11



First Activity from Second IP:

2026-04-20T03:35:44Z

This IP was later used to: create access keys, attempt log tampering and maintain long term access

3. Obfuscated Data Analysis

Two Base64-encoded strings were identified and decoded.

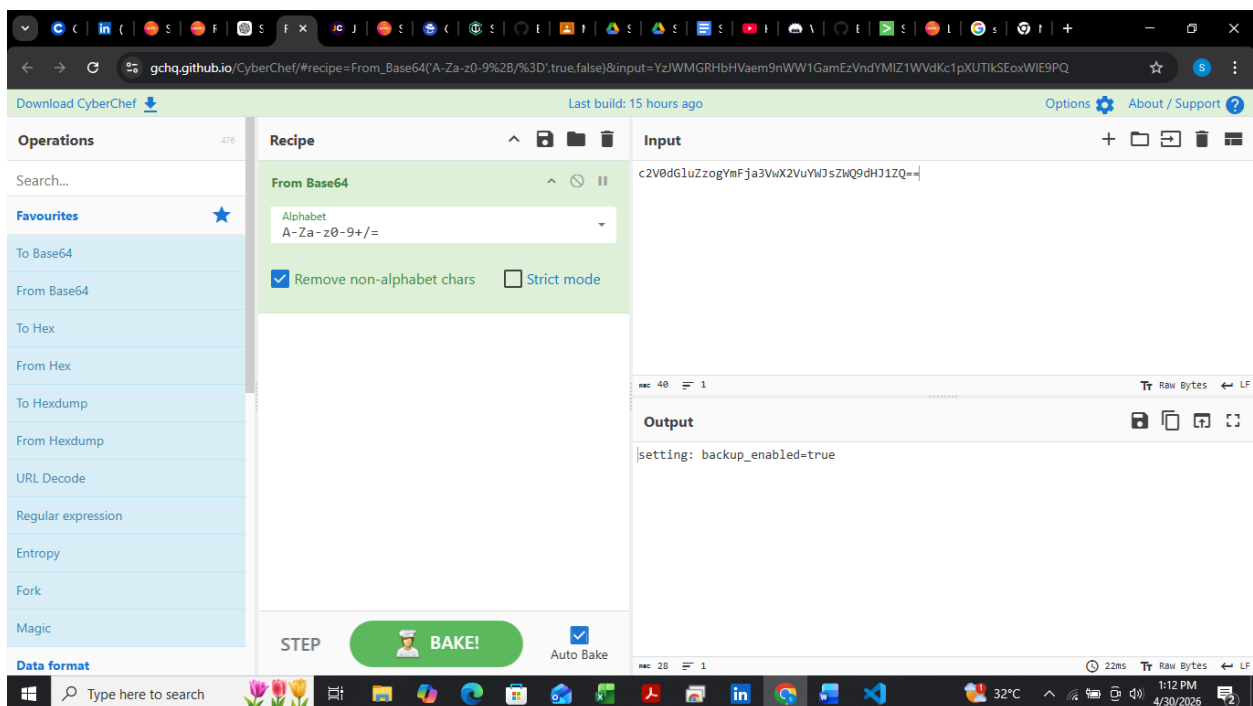
Legitimate Configuration Data:

c2V0dGluZzogYmFja3VwX2VuYWJsZWQ9dHJ1ZQ==

```
50 {
51   "eventName": "DescribeSecurityGroups",
52   "awsRegion": "us-east-1",
53   "sourceIPAddress": "172.31.45.12",
54   "userAgent": "aws-internal/3"
55 },
56 {
57   "eventName": "PutObject",
58   "awsRegion": "us-east-1",
59   "sourceIPAddress": "10.0.1.55",
60   "userAgent": "DevOps-Deployer/1.0",
61   "requestParameters": {
62     "bucketName": "app-configs",
63     "key": "prod_config.ini",
64     "content": "c2V0dGluZzogYmFja3VwX2VuYVJ3ZWQ9dHJ1ZQ=="
65   }
66 },
67 {
68   "eventTime": "2026-04-20T01:25:44Z",
69   "eventSource": "iam.amazonaws.com",
70   "eventName": "GetAccountSummary",
71   "awsRegion": "us-east-1",
72   "sourceIPAddress": "190.45.112.3",
73   "userAgent": "aws-cli/2.0.30",
74   "userIdentity": {
75     "accessKeyId": "AKIA_DEV_PROD_9982"
76   }
77 },
78 {
79   "eventTime": "2026-04-20T01:30:12Z",
80 }
```

Decoded:

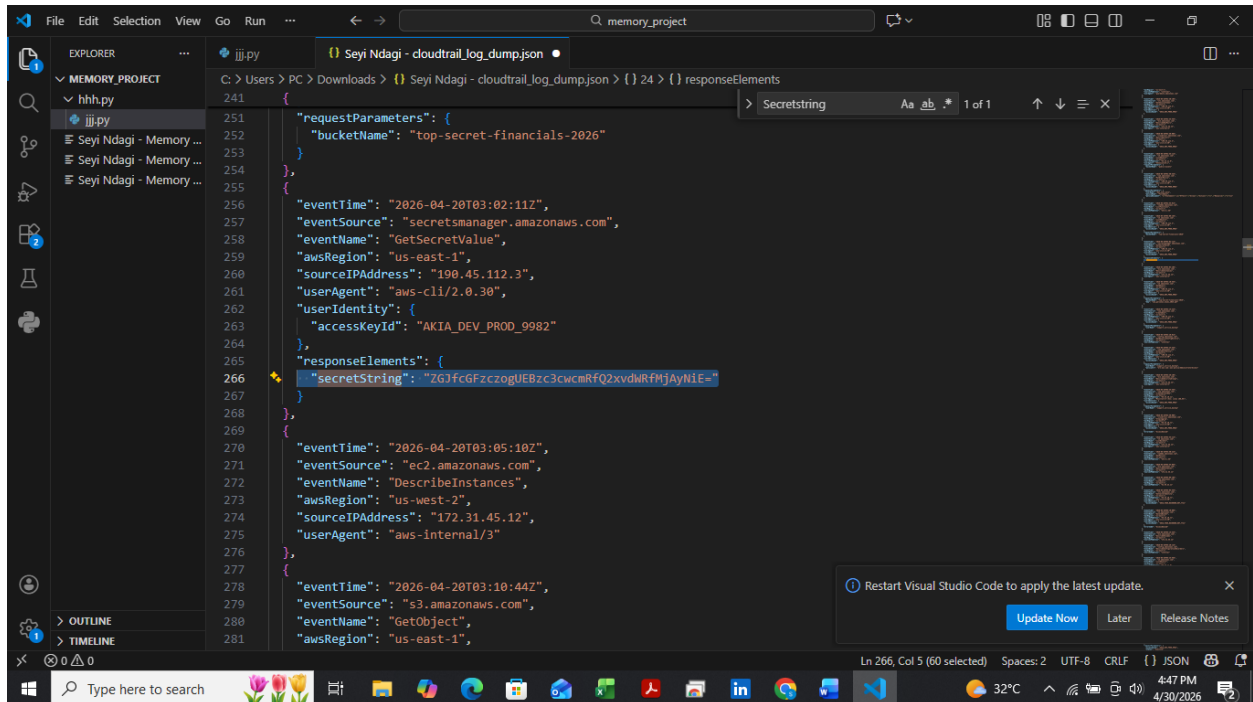
setting: backup_enabled=true



This represents normal application configuration.

Sensitive Credential Exposure:

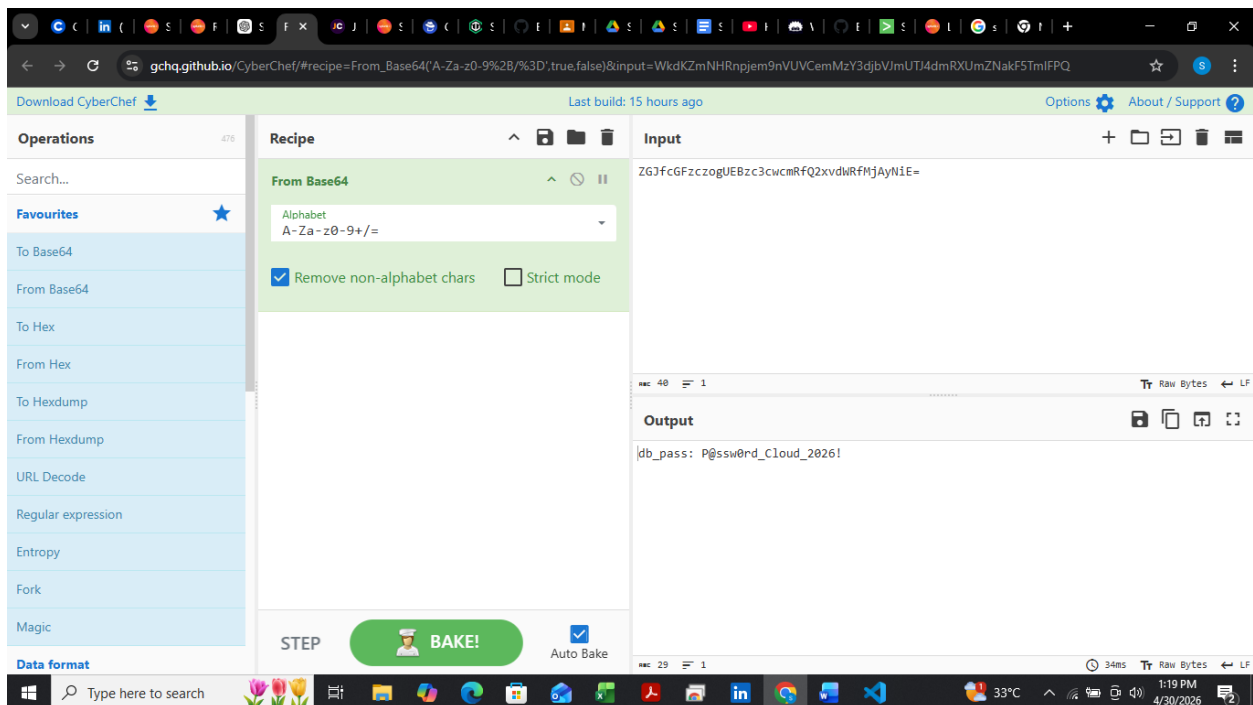
ZGJfcGFzczogUEBzc3cwcmRfQ2xvdWRfMjAyNiE=



```
251 {
252   "requestParameters": {
253     "bucketName": "top-secret-financials-2026"
254   },
255   "eventTime": "2026-04-20T03:02:11Z",
256   "eventSource": "secretsmanager.amazonaws.com",
257   "eventName": "GetSecretValue",
258   "awsRegion": "us-east-1",
259   "sourceIPAddress": "190.45.112.3",
260   "userAgent": "aws-cli/2.0.30",
261   "userIdentity": {
262     "accessKeyId": "AKIA_DEV_PROD_9982"
263   },
264   "responseElements": {
265     "secretString": "ZGJfcGFzczogUEBzc3cwcmRfQ2xvdWRfMjAyNiE="
266   },
267 },
268 {
269   "eventTime": "2026-04-20T03:05:10Z",
270   "eventSource": "ec2.amazonaws.com",
271   "eventName": "DescribeInstances",
272   "awsRegion": "us-west-2",
273   "sourceIPAddress": "172.31.45.12",
274   "userAgent": "aws-internal/3"
275 },
276 {
277   "eventTime": "2026-04-20T03:10:44Z",
278   "eventSource": "s3.amazonaws.com",
279   "eventName": "GetObject",
280   "awsRegion": "us-east-1",
281 }
```

Decoded:

db_pass: P@ssw0rd_Cloud_2026!



This reveals a **database password**, indicating credential leakage.

4. Privilege Escalation

The attacker granted full administrative access using:

API Call: PutUserPolicy

Timestamp: 2026-04-20T02:50:33Z

This action allowed unrestricted control over AWS resources.

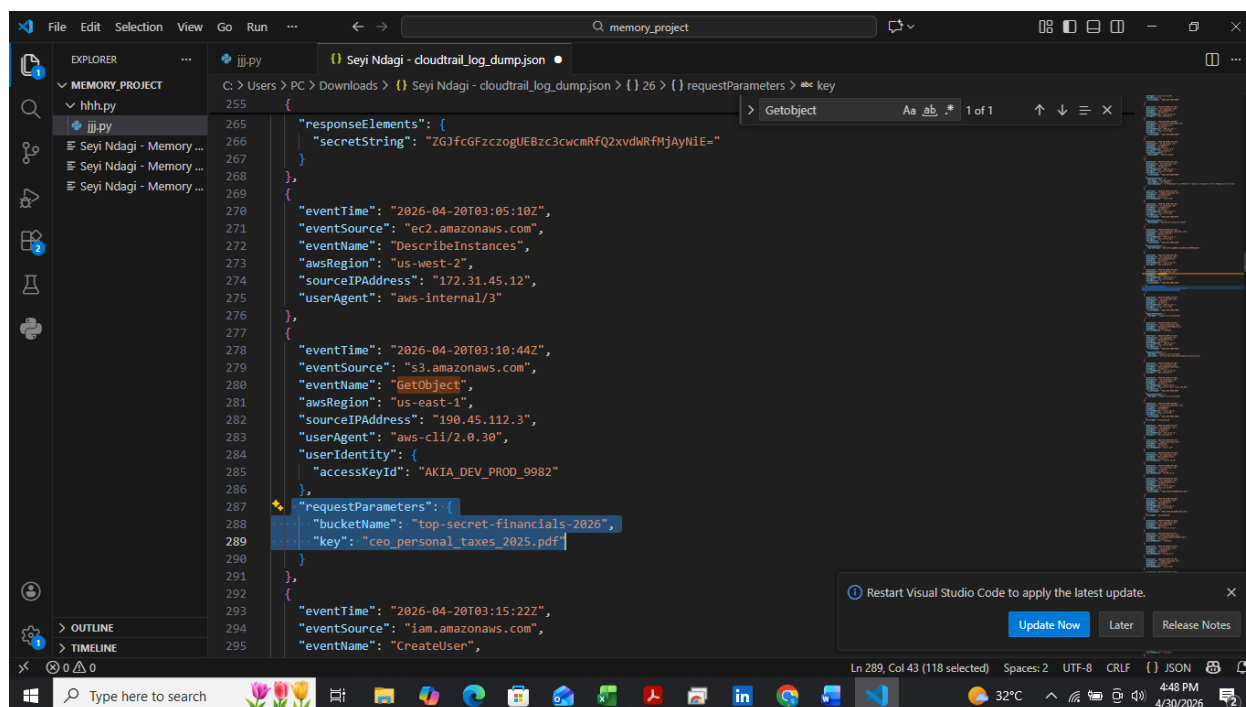
5. Data Exfiltration

The attacker accessed sensitive data from an S3 bucket:

Bucket Name: top-secret-financials-2026

File Name: ceo_personal_taxes_2025.pdf

This indicates targeted exfiltration of high-value financial data.



```
255 {
256   "responseElements": {
257     "secretString": "ZGJfcGFzc2ogUEBzc3cwcmlRfQ2xvdwRfMjAyN1E="
258   },
259   "eventTime": "2026-04-20T03:05:10Z",
260   "eventSource": "ec2.amazonaws.com",
261   "eventName": "DescribeInstances",
262   "awsRegion": "us-west-2",
263   "sourceIPAddress": "172.31.45.12",
264   "userAgent": "aws-internal/3"
265 },
266 {
267   "eventTime": "2026-04-20T03:10:44Z",
268   "eventSource": "s3.amazonaws.com",
269   "eventName": "GetObject",
270   "awsRegion": "us-east-1",
271   "sourceIPAddress": "190.45.112.3",
272   "userAgent": "aws-cli/2.0.30",
273   "userIdentity": {
274     "accessKeyId": "AKIA_DEV_PROD_9982"
275   },
276   "requestParameters": {
277     "bucketName": "top-secret-financials-2026",
278     "key": "ceo_personal_taxes_2025.pdf"
279   },
280 },
281 {
282   "eventTime": "2026-04-20T03:15:22Z",
283   "eventSource": "iam.amazonaws.com",
284   "eventName": "CreateUser",
285 }
```

6. Persistence Mechanism

Backdoor User Created:

support_service_backup

```
279 {
280   "eventSource": "s3.amazonaws.com",
281   "eventName": "GetObject",
282   "awsRegion": "us-east-1",
283   "sourceIPAddress": "190.45.112.3",
284   "userAgent": "aws-cli/2.0.30",
285   "userIdentity": {
286     "accessKeyId": "AKIA_DEV_PROD_9982"
287   },
288   "requestParameters": {
289     "bucketName": "top-secret-financials-2026",
290     "key": "ceo_personal_taxes_2025.pdf"
291   },
292 },
293 {
294   "eventTime": "2026-04-20T03:15:22Z",
295   "eventSource": "iam.amazonaws.com",
296   "eventName": "CreateUser",
297   "awsRegion": "us-east-1",
298   "sourceIPAddress": "190.45.112.3",
299   "userAgent": "aws-cli/2.0.30",
300   "userIdentity": {
301     "accessKeyId": "AKIA_DEV_PROD_9982"
302   },
303   "requestParameters": {
304     "userName": "support_service_backup"
305   },
306 },
307 {
308   "eventTime": "2026-04-20T03:20:01Z",
309   "eventSource": "config.amazonaws.com",
310   "eventName": "GetResourceConfigHistory",
```

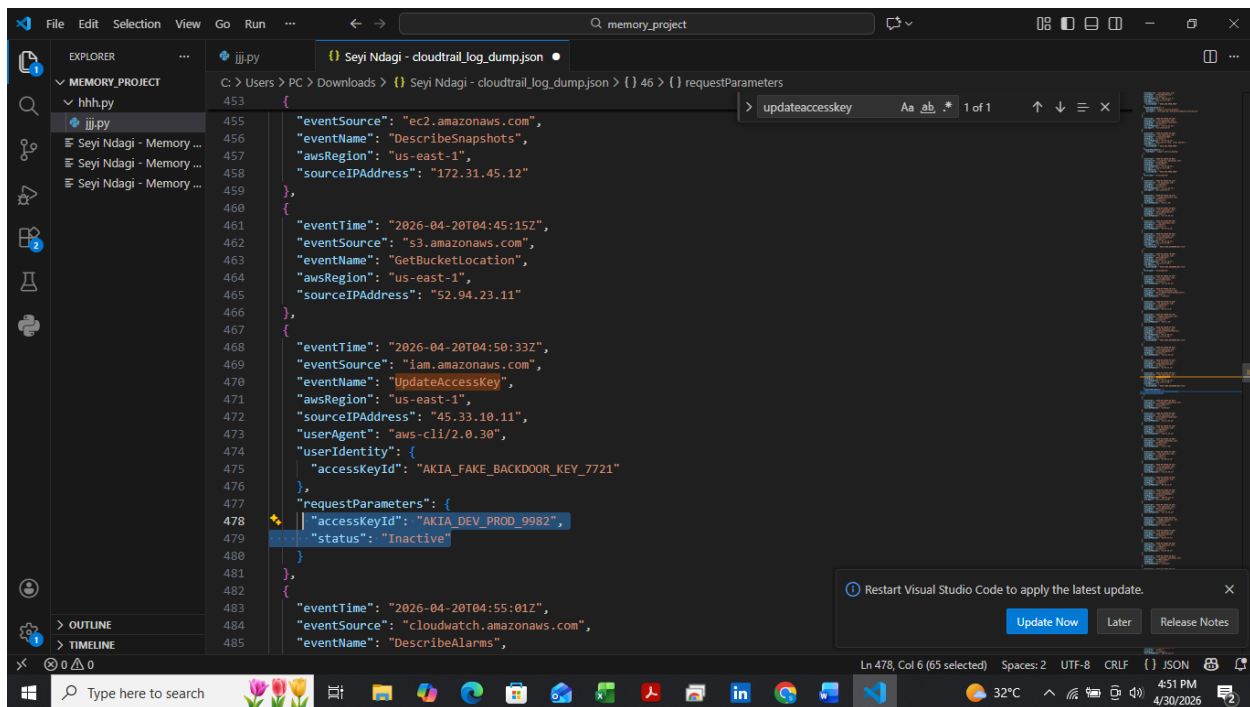
The attacker:

Created a new IAM user, assigned administrative privileges and generated new access keys

7. Evasion Technique

The attacker performed the following action:

UpdateAccessKey: set original key to Inactive



Why this matters:

It legitimate user access, reduces suspicion on compromised key and shifts control to attacker created backdoor.

Attack Timeline Summary

1. Attacker logs in using stolen credentials
2. Performs reconnaissance (listing resources)
3. Escalates privileges to full admin
4. Extracts sensitive credentials and data
5. Downloads confidential file from S3
6. Creates backdoor user for persistence
7. Disables original access key to evade detection

Conclusion

The investigation reveals a complete cloud compromise involving credential theft, privilege escalation, data exfiltration, and persistence.

The attacker demonstrated advanced techniques including:

use of stolen AWS credentials

obfuscation via Base64 encoding

creation of backdoor identities

operational evasion tactics

Recommendations

Rotate and secure all access keys

enforce least privilege IAM policies

enable multi-factor authentication (MFA)

monitor unusual API activity via CloudTrail

restrict access based on IP and behavior

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